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Description - The Full Size PDU is a two-bay cabinet, and the Compact PDU is a single-bay cabinet. Procedures for the Full Size PDU do not apply to the Compact PDU.

This section checks out the control circuitry in the Full Size PDU, including Mode Control / Mode Sequencing, Emergency Stop operation, Main Transformer Over temperature Protection, and Door Interlock Operation.

1- INITIAL SET UP

1. Verify that main disconnect is off, locked, and tagged.
2. Circuit breakers:
Nonregulated PDU: Close 400A circuit breaker (PD1A4CB18).
PDU with PowerTECH: Close main input breaker on regulator.
3. Verify that all other circuit breakers are off (open).



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN WHEN ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK, DISCONNECT POWER FROM THE PDU BEFORE PERFORMING THE FOLLOWING PROCEDURE.

4. Close main disconnect.
5. Reset Emergency Off.
6. Verify that Transient Protection Nominal neon lights (DS1, DS2, DS3) are on. No other lights should be on, and no fans should be running.
7. Close PDU power circuit breaker (PD1A5CB10).
8. Verify PDU control board cycles are in Full Off mode. No contactors should be energized (K1 through K12) and Full Off amber light should be on.
9. Fan verification
Non-regulated PDU: Verify that all six fans are running. Lower three fans should be drawing air into cabinet, upper three should be exhausting air from cabinet.
PDU with PowerTECH: Verify that all three fans are running.

2- MODE CONTROL / MODE SEQUENCING

The order of checks performed in this procedure efficiently covers all modes of operation.

1. Visually verify that contactors K1 through K12 energize in sequences noted below for each mode transition specified. Also, verify that correct indicator lights energize for each mode. Refer to Table 1 below, Table 2 under Table 1, and Illustration L2699A under Table 2.

TABLE 1
SUBSYSTEM LOGICAL GROUPING

Group	Component	Contactor	Feeder	Breaker
A	RF / Pen (Spectro Broadband Amp functions) Cabinet (MR1)	K10	1	CB17
	Gradient Driver (High Power Module + GAP) Cabinet (MR3)	K9	4	CB7
	Gradient Driver (Low Power Module Cabinet (MR3)	K11	5	CB6
	GRAM Cabinet (MR7)	K5	2	CB5
B	Shim Power Supply Cabinet (MG4 / MM4 / MS3)	K8	6	CB16
	Main Magnet Power Supply Cabinet (MG5 / MM5 / MS2)	K7	7	CB15
C	None			
D	System Control Cabinet (MR2)	K3	10	CB11
	Operator Work Space (OW1)	K12	15	CB8
E	Spare (Customer Options)	K1	9	CB1
	Spare	K2	12	CB4
	Heat Exchanger	K2	11	CB3
	Multiformat Unit (MF1) / Laser Camera (LC1) (optional)	K2	8	CB2
	B0 for Mobiles	K4	13	CB13
F	PDU Power (PD1)	N/A	17	CB10
	Service Outlets	N/A	16	CB9

TABLE 2
PDU OPERATING MODES AND GROUP STATUS

MODE	GROUP/STATUS					
	A	B	C	D	E	F
#1 Emergency Off	OFF	OFF	OFF	OFF	OFF	OFF
#2 Full Off	OFF	OFF	OFF	OFF	OFF	ON
#3 Emergency Stop	OFF	OFF	OFF	ON	ON	ON
#4 Overnight	OFF	ON	ON	ON	ON	ON
#5 Full On	ON	ON	ON	ON	ON	ON

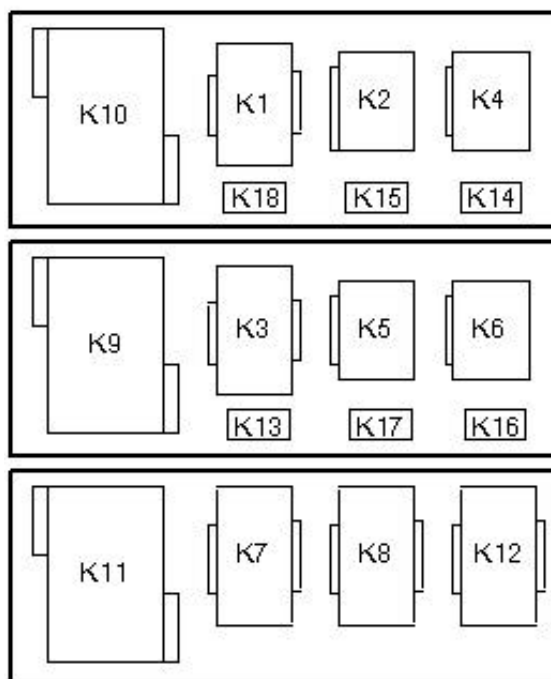


ILLUSTRATION L2699A
CONTACTOR LOCATIONS

Allow sequencing to complete between each cycle.

FULL OFF → FULL ON

- K1, K2, K3, K4 energize
- Wait 5 seconds
- K5, K6 energize

- d. Wait 5 seconds
- e. K7, K8, K9, K10, K11, K12 energize

FULL ON → EMERGENCY STOP

- a. (K1 through K12 initially energized)
- b. K6, K7, K8, K9, K10, K11, K12 de-energize

EMERGENCY STOP → OVERNIGHT

- a. (K1 through K5 initially energized)
- b. K6 energizes
- c. Wait 5 seconds
- d. K7, K8 energize

OVERNIGHT → FULL ON

- a. (K1 through K8 initially energized)
- b. K9, K10, K11, K12 energize

FULL ON → FULL OFF

- a. (All contactors initially energized)
- b. K7, K8, K9, K10, K11, K12 de-energize
- c. Wait 5 seconds
- d. K5, K6 de-energize
- e. Wait 5 seconds
- f. K1, K2, K3, K4 de-energize

FULL OFF → OVERNIGHT

- a. K1, K2, K3, K4 energize
- b. Wait 5 seconds
- c. K5, K6 energize
- d. Wait 5 seconds
- e. K7, K8 energize

OVERNIGHT → EMERGENCY STOP

- a. (K1 through K8 initially energized)
- b. K6, K7, K8 de-energize

EMERGENCY STOP → FULL ON

- a. (K1 through K5 initially energized)
- b. K6 energizes
- c. Wait 5 seconds
- d. K7, K8, K9, K10, K11, K12 energize

FULL ON → OVERNIGHT

- a. (All contactors initially energized)
- b. K9, K10, K11, K12 de-energize

OVERNIGHT → FULL OFF

- a. (K1 through K8 initially energized)
- b. K7, K8 de-energize
- c. Wait 5 seconds
- d. K5, K6 de-energize
- e. Wait 5 seconds
- f. K1, K2, K3, K4 de-energize

FULL OFF → EMERGENCY STOP

- a. Verify that the mode does not change, and that no contactors are energized. This mode transition is not allowed.
- b. Next, select Full On mode, then, when sequencing is complete, select Emergency Stop mode.

EMERGENCY STOP → FULL OFF

- a. (K1 through K5 initially energized from previous step)
- b. K5 de-energizes
- c. Wait 5 seconds
- d. K1, K2, K3, K4 de-energize

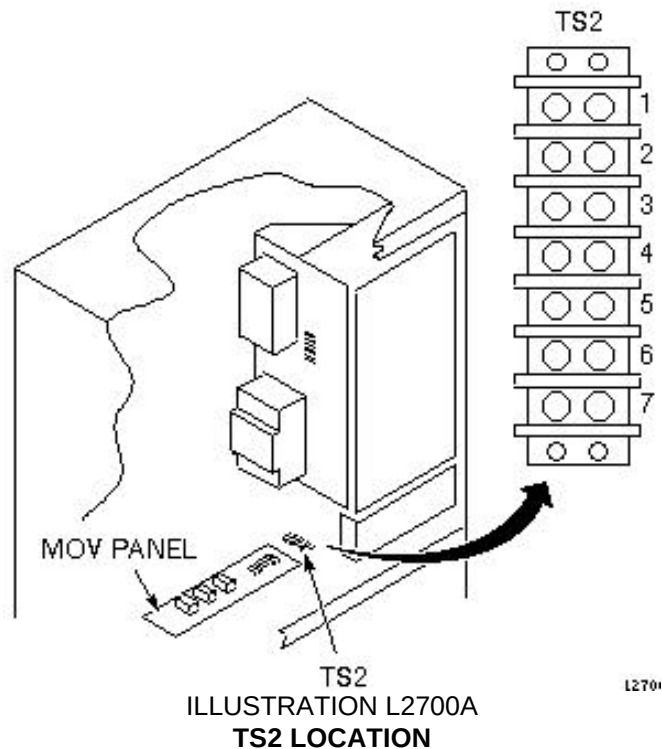
3- EMERGENCY STOP CIRCUIT

1. Select Full On mode and wait for completion of sequencing.
2. Remove the jumper or cable on J2 of the PDU control board (PD1A1A2J2).
3. Verify that PDU enters Emergency Stop mode.
4. Reconnect the jumper or cable and select Full On.

4- PDU OVERTEMPERATURE PROTECTION UNIT

FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES EXIST ON THE MOV PANEL IMMEDIATELY NEXT TO TS2. EXERCISE THE UTMOST CAUTION DURING THE NEXT TWO STEPS. FAILURE TO DO SO COULD CAUSE SEVERE INJURY OR DEATH.

1. Connect a jumper between TS2-1 and TS2-2. See Illustration L2700A below.



2. Verify that white Over temp indicator (PD1A1A2DS9) on front of PDU door (PD1A1) illuminates.
3. Remove jumper and verify indicator turns off.
4. Connect a jumper between TS2-2 and TS2-3. See Illustration L2700A above.
5. Verify that main disconnect trips (Emergency Off enabled) and all power to PDU is off.
6. Verify that Temp Exceeded indicator located on A5 panel has set (white).
7. Remove jumper and reset main disconnect.
8. Select Full On mode and verify that Temp Exceeded indicator has reset (black).
9. Verify that all elapsed time indicators, if present, are running when in Full On mode.

5- DOOR INTERLOCK CIRCUIT

1. Open top doors of PDU.
2. Press door interlock switch. See Illustration L2701A below.

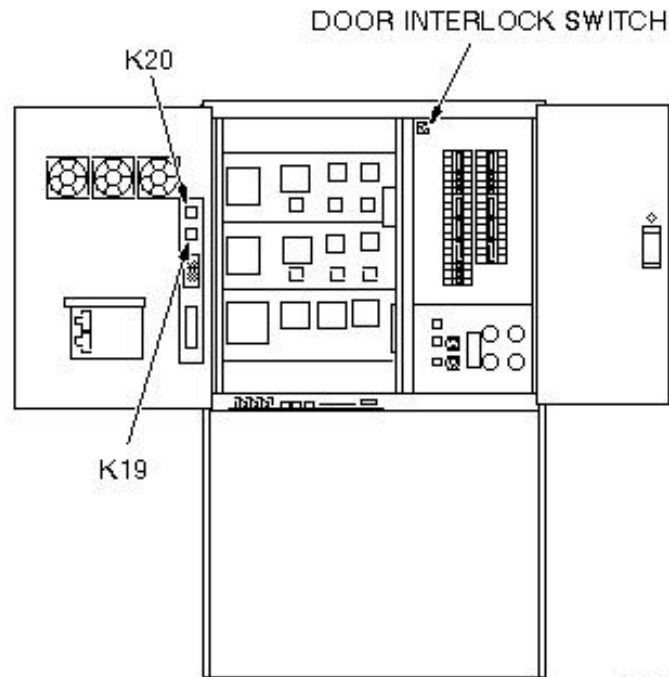


ILLUSTRATION L2701A
DOOR INTERLOCK COMPONENTS

3. Verify that relays K19 and K20 energize.
4. Release switch, and verify that K19 and K20 de-energize.

6- EMERGENCY STOP BUTTONS

6-1 Description

There are five EMERGENCY STOP buttons in the system:

- Two on the magnet enclosure (MG2)
- One on the operator workspace keyboard
- One on the PDU (PD1)
- One on the remote power panel (RPP)

The purpose of the Emergency Stop mode is to remove power in the patient vicinity.

6-2 Initial Conditions

Signa software does not have to be up since this is a hardware function. If Signa software is up when EMERGENCY STOP is pressed, a message appears on the monitor indicating that EMERGENCY STOP has been pressed.

1. Verify that all cabinets are powered up (Full On mode at PDU).
2. Press EMERGENCY STOP button at operator workspace keyboard and verify that power is removed from the following cabinets:
 - RF / Pen Power Cabinet (MR1)
 - Gradient Driver Cabinet (MR3)
 - GRAM Cabinet (MR7)
 - Shim Power Supply Cabinet (MG4)
 - Main Magnet Power Supply Cabinet (MG5).
3. Verify that EMERGENCY STOP button illuminates at PDU.
4. Press Full On at PDU to restore power.
5. Repeat steps 1 through 4 for the other four EMERGENCY STOP buttons.

Note

When test is complete, press the green START button on front of shim power supply cabinet to turn the Oxford shim power supply back on.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 17, 1998	M. Whitlow	Initial Conversion from Toolbook to Word.